CollapseCore Campus Network: Updated Project Overview

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Summary

CollapseCore is a working campus network that now carries wireless, a company web portal and guest isolation on top of the original redundant core, zone firewall and RADIUS login. The lab file has 45 devices: two collapsed-core multilayer switches, two floor switches, a zone-based firewall, a wireless controller with two access points, and eight servers.

Since the first overview, six things were added or fixed:

- A WLC-3504 with two lightweight APs broadcasting `ITSupport-Client`, `ITSupport-Executive` and `ITSupport-Guest`, each landing in its own VLAN.
- AP uplinks rebuilt as trunks with native VLAN 999, and FlexConnect local switching turned on for all three WLANs.
- A guest isolation ACL (`GUEST-IN`) applied to the Guest VLAN on both cores.
- A management DHCP pool sized to the usable range, with the controller address handed to the APs.
- A new web server, ITSupport-Web (192.168.1.9), allowed through the firewall and hosting the company portal.
- A written troubleshooting log of what failed and why.

The lab is verified for wireless: clients on each SSID receive an address from that SSID's subnet. Open items are listed in the last section.

Topology and addressing

The network is a collapsed-core campus: two multilayer core switches, two access switches, a zone-based firewall in front of a server VLAN, and a wireless controller with two lightweight access points.

Devices

Role	Device	Notes
Core (primary)	BLD1-CORE-SW1	STP root, HSRP active
Core (secondary)	BLD2-CORE-SW1	STP secondary, HSRP standby
Access	FL1	User floor, also hosts both APs
Access	EXEC-SW	Executive floor
Firewall	Zone-based firewall	INSIDE, SERVERS and OUTSIDE zones, OSPF to the cores
Wireless	WLC-3504	Management address 128.137.234.70
Wireless	2 x LAP-PT	Join the controller through DHCP option, trunk ports

VLANs and subnets

VLAN	Name	Subnet	Gateway
30	Client	128.137.232.0/24	.254 (HSRP)
40	Voice	128.137.233.0/24	.254 (HSRP)
70	Printer	128.137.234.0/27	.30 (HSRP)
80	Executive	128.137.234.32/27	.62 (HSRP)
100	Guest	128.137.234.96/27	.126 (HSRP)
200	Servers	192.168.1.0/24	Firewall
999	Management	128.137.234.64/27	.94 (HSRP)

Servers (VLAN 200)

Service	Address
DHCP	192.168.1.2
DNS	192.168.1.3

Service	Address
Web (ITSupport portal)	192.168.1.4
Web (ITSupport staff console)	192.168.1.9
Email (SMTP/POP3)	192.168.1.5
AAA (RADIUS)	192.168.1.6
NTP	192.168.1.7
TFTP	192.168.1.8

Switching and redundancy

Two multilayer switches act as one collapsed core, and every floor switch connects to both, so one failure does not take a VLAN down.

- **Spanning tree:** PVST, with BLD1-CORE-SW1 at priority 4096 (root) and BLD2-CORE-SW1 at 8192 for VLANs 30, 40, 70, 80, 90, 100, 200, 600 and 999.
- **First-hop redundancy:** HSRPv2 on every routed VLAN. BLD1 has priority 105 with preempt and is active. Each core tracks its firewall uplink (Gi1/1/1 on BLD1, Gi1/1/2 on BLD2), so a lost uplink hands the gateway to the other core.
- **Uplinks:** FL1 and EXEC-SW are each dual-homed (Gi0/1 and Gi0/2) to the two cores. BLD1 and BLD2 are joined by a trunk on Gi1/1/4.
- **Floor-to-floor link:** FL1 and EXEC-SW share an LACP port-channel on Fa0/23 and Fa0/24. Spanning tree blocks it for VLAN 999 by design.
- **Trunks:** native VLAN 90 on all inter-switch links, with only the VLANs in use allowed.
- **Access ports:** PortFast, BPDU guard, sticky port security (max 5), voice VLAN 40 and QoS trust on user ports.
- **DHCP:** every routed VLAN has `ip helper-address 192.168.1.2` on both cores, so one server serves all subnets.
- **Management:** VLAN 999 (128.137.234.64/27) with HSRP address .94. Switches use SSH version 2 only on the VTY lines, with RADIUS login and a local fallback, and the console uses a separate local login.

Routing and security

- **Routing:** OSPF area 0 between both cores and the firewall. VLAN 200 (servers) is reached through the firewall, which is its gateway.
- **Zone-based firewall:** three zones. INSIDE (all user VLANs) may reach SERVERS and OUTSIDE. OUTSIDE may not initiate toward INSIDE or SERVERS.
- **Server access list (ACL-IN-TO-SRV):** permits only the services each server provides: web on 192.168.1.4 and 192.168.1.9, DNS, DHCP, SMTP/POP3, and RADIUS (UDP 1645 and 1646). The missing entry for the second web server (.9) was why the new site timed out while ping worked; adding it fixed access.
- **Guest isolation:** access list GUEST-IN on VLAN 100, applied on both cores, allows DHCP, DNS and internet traffic and blocks all internal networks.
- **AAA:** RADIUS on 192.168.1.6 authenticates switch logins and the Client and Executive SSIDs (802.1X). Local accounts remain as a fallback.
- **Device hardening:** SSH version 2 only on VTY lines, separate local console login, authenticated NTP, unused ports shut down, sticky port security on access ports.

Wireless build

A WLC-3504 controls two LAP-PT access points that broadcast three SSIDs, each mapped to its own VLAN and subnet.

SSID	Audience	Security	VLAN	Subnet
ITSupport-Client	Staff	WPA2-Enterprise (802.1X)	30	128.137.232.0/24
ITSupport-Executive	Leadership	WPA2-Enterprise (802.1X)	80	128.137.234.32/27
ITSupport-Guest	Visitors	WPA2-Personal (PSK)	100	128.137.234.96/27

How it is built

- The controller management interface is untagged, so the switch trunk toward it uses native VLAN 999. Each SSID has a dynamic interface carrying its VLAN ID.
- The APs sit on trunk ports (native 999, allowed 30, 80, 100, 999) and find the controller through the WLC Address option in the management DHCP pool.

- FlexConnect Local Switching is turned on. This was the fix that finally put each SSID's clients on its own VLAN instead of the management range.

Result: clients on each SSID now receive an address from that SSID's own subnet, and the Guest SSID reaches only DNS and the internet.

Packet Tracer quirks to remember

- Secured SSIDs do not appear under Connect. A profile with the exact SSID name and security type must be created on the Profiles tab first.
- After cable or AP changes, power-cycle the AP from its Physical tab. Stale AP entries on the controller are harmless.

Web services

Two web servers run the ITSupport mock company sites.

- **192.168.1.4:** public-facing ITSupport portal (services, Wi-Fi table, network directory, help center) in plain HTML, styled for Packet Tracer's browser.
- **192.168.1.9:** ticket portal (`index.html`) and staff console (`admin.html`). Tickets are stored in the browser, so they only appear on the PC that created them.

Packet Tracer versus a real server. Packet Tracer's HTTP server serves static files only and does not run JavaScript backends or JSON APIs, so the lab uses the static versions of the pages. A zero-dependency Node server (`server.js` with a `tickets.json` store and an admin PIN) is included outside the lab; with it, a ticket submitted from any PC shows on every PC in the staff console. The full set of scripts is packaged as `itsupport.zip`.

Troubleshooting log

Problem	Root cause	Fix
APs registered but SSIDs showed nothing on PCs	Secured SSIDs are hidden from Connect in Packet Tracer	Create a profile per SSID on the Profiles tab
Clients got management addresses (.84-.87)	FlexConnect Local Switching was off	Enable Local Switching so each SSID uses its VLAN

Problem	Root cause	Fix
AP lost its connection or never joined	AP port cabling and stale port security entries	Fix cabling, clear port security, power-cycle the AP
WLC unreachable from the switch side	Management is untagged	Use native VLAN 999 on the controller trunk
New web server 192.168.1.9 pinged but timed out on HTTP	Missing entry in the server access list	Add .9 to ACL-IN-TO-SRV
Tickets only visible on the submitting PC	Browser-local storage	Use the Node ticket server outside Packet Tracer
Guest client received an address outside its subnet	Pool and VLAN mismatch before Local Switching was on	Resolved by the FlexConnect change

Lessons

- Packet Tracer's wireless controller differs from real Cisco gear; follow its own lab behavior rather than production guidance.
- When ping works but HTTP does not, check the firewall access list for that exact host and port.
- Change one thing, then verify with a client, instead of stacking changes.

Open items and next steps

- Enlarge the Guest DHCP pool: it holds only .98 to .103 (six leases); target .99 to .123.
- Move the VIP DHCP pool start to .35, since .34 is BLD2's VLAN 80 interface address.
- Verify the RADIUS shared secret matches on both cores and the AAA server's client entries.
- Unplug Laptop6 from the WLC Gi5 port and clear the stale sticky entries on FL1 Fa0/21.
- Allow NTP through the firewall so authenticated time sync works end to end.
- Clean up the leftover call-manager (CME) configuration and re-register the second phone.
- Update the older runbook document to match this build.
- Add a Wireless section to the original CollapseCore overview.

- State the portfolio audience so the write-ups can be tuned for it.